

Total No. of Questions : 12]

SEAT No. :

P3109

[5461]-146

[Total No. of Pages : 4

B.E. (Mechanical)

INDUSTRIAL ENGINEERING

(402049C) (2012 Pattern) (Semester - II) (Elective - III) (End Sem.)

Time : 2 ½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8, Q9 or Q10 or Q11 or Q12.
- 2) Answer should be written in one answer book.
- 3) Neat diagrams must be drawn wherever necessary.
- 4) Figures to the right side indicate full marks.
- 5) Assume suitable data if necessary.

Q1) a) Explain with neat flow chart types of organisation. **[4]**

- b) A manufacturing unit is producing 8,000 products by employing 50 workers at 8 hrs./day for 25 days in a month. If company increases this quantity to 9,600 products by hiring 10 additional workers at 8 hrs./day for 25 days in a month. Is there increase in labor productivity? What will be the increase in labor productivity if company increases quantity to 10,000 products without hiring additional workers? **[4]**

OR

Q2) a) Discuss the scope and any four objectives of industrial engineering. **[4]**

- b) A company produces 1500 kg of tablets by consuming 2000 kg mix of raw material for a particular period. For the next period, the output is 3200 kg by consuming 4265 kg of the raw material and for the third period, the output is increased to 3500 kg by consuming 4200 kg of raw material. Comment on increase or decrease of productivity for second and third period. **[4]**

Q3) State the various steps of method study and explain in detail critical examination of methods. **[6]**

OR

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Q4) Explain principles of motion economy related to use of human body, arrangement of workplace and design of tools and equipment. [6]

Q5) The two steps in preparing chocolate candy bars are molding and packaging. Personal fatigue and delay allowances are set at 12%. The molding machine operator is rated at 90% and the packer is rated at 110%. Observed times per batch are given below. Determine the normal and standard time for both task. [6]

	observed time in minutes			
Task	1	2	3	4
Molding	26	30	29	31
Packing	45	50	35	30

OR

Q6) A group of 10 workers working for 8 hours per day produced 320 pieces of component. It was observed that 15% of the total available time workers were idle and for remaining time they worked at an average performance of 80%. Calculate standard time for the Job assuming operations to be completed manually and workers are entitled for 12% relaxation allowances. [6]

Q7) a) Explain in brief various functions of production planning and control. [8]
 b) Historical data on the sale of mobile phones of a company for the year 2011 to 2017 is given below. Using the regression analysis establish the trend values and extrapolate for the year 2018 ignoring the effect of seasonal and random fluctuations. What is the forecast for the year 2018. [9]

Year	2011	2012	2013	2014	2015	2016	2017
Sales	3286	4751	5867	4580	5020	8444	11072

OR

Q8) a) Explain the concept of supply chain management and supply chain network. Describe any one order control strategy in detail. [8]

b) Historical data on the sale of lubrication pumps for the last 12 years is given below. By the method of three yearly moving average establish the trend values and forecast demand for 13th year. If actual demand for 13th year is 520 nos. what shall be the forecast for 14th year? [9]

Year	1	2	3	4	5	6	7	8	9	10	11	12
Sales												
Nos.	332	344	328	336	370	408	420	432	428	450	498	540

Q9) a) Explain in detail factors affecting selection of location for the industrial plant. [8]

b) A company buys an item in lots of 500 units which is a three months requirement. The cost per unit is Rs.90 and the ordering cost in Rs.180 per batch order. The inventory carrying cost is estimated at 20% of the average inventory investment. What is the annual total cost of existing inventory policy? How much money can be saved from economic order quantity purchase? [9]

OR

Q10)a) What are different principles of material handling? Explain different material handling equipment's with their applications. [8]

b) A company spends Rs. 34,000/- on its purchasing activity and Rs.67,200/- for maintenance of inventory of Rs. 4,20,000/- annually. Around 850 orders are placed every year to replenish stocks of the various items. One of the item whose annual consumption is 9600 nos. costing Rs.30/- each is brought by the company based on staggered deliveries. How frequently should the company receive staggered deliveries and in what quantities? What is the corresponding annual total cost for this item. [9]

Q11)a) An industry manufacturing small capacity motors has the cost structure as follows

Material cost :Rs.50

Labor cost: Rs. 80

Variable overhead : 75% of labour cost.

Fixed overhead : Rs. 2,40,000/annum.

Sales price: Rs. 230/motor

Determine the number of motors to be manufactured to break-even.

Number of motors to be sold to make a profit of Rs.1,00,000/-

Number of motors to be sold to break even if price is reduced by

Rs. 15/ motor. [8]

- b) What do you understand by industrial safety? What are different safety issues and general safety rules in industry? [8]

OR

Q12)a) Explain break-even analysis with neat chart? What are assumptions made in break-even analysis? What are different applications of break-even analysis? [8]

- b) Explain in brief different functions of human resource department. Comment on concept of key result areas and performance appraisal of employee. [8]

